Chapter 6 — Chemical Energetics

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Enthalpy change and energy profile diagrams

Q.1 Explain enthalpy change, exothermic and endothermic reactions, and energy profile diagrams.

Enthalpy change, ΔH, is the heat change of a reaction at constant pressure. It equals the enthalpy of products minus the enthalpy of reactants. In an exothermic reaction, heat is released, products have lower enthalpy and ΔH is negative. In an endothermic reaction, heat is absorbed, products have higher enthalpy and ΔH is positive.

An energy profile diagram shows the energy pathway from reactants to products. The peak represents the activation energy. Exothermic profiles end lower than they start, while endothermic profiles end higher. Catalysts lower the activation energy but do not change ΔH.

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Important Conceptual Short Questions — Topic 1

1. Why is ΔH negative for combustion?

Combustion releases heat because the products, usually CO_2 and H_2O, have lower enthalpy than the reactants. Therefore ΔH is negative.

2. How does an energy profile diagram show activation energy?

Activation energy is the energy gap between the reactants and the top of the energy barrier. It is the minimum energy needed for an effective reaction.

3. Why does a catalyst not change the enthalpy change of a reaction?

A catalyst provides an alternative pathway with lower activation energy, but the initial and final enthalpy levels remain the same. Therefore ΔH is unchanged.

4. How can bond breaking and bond making explain exothermic and endothermic changes?

Bond breaking absorbs energy and bond making releases energy. If bond making releases more energy than bond breaking absorbs, the reaction is exothermic; otherwise it is endothermic.

Topic 2 — Standard enthalpy changes and Hess’s law

Q.2 Define standard enthalpy changes and use Hess’s law to calculate enthalpy changes.

Standard enthalpy changes are measured under standard conditions: 298 K, 1 atm pressure and 1 mol dm^{-3} solutions. Important types include enthalpy of reaction, combustion, formation, atomization and neutralization. The standard enthalpy of formation of an element in its standard state is zero. For strong acid-strong base neutralization, ΔH is about –57.1 kJ mol^{-1}.

Hess's law states that the total enthalpy change is independent of the pathway, provided initial and final states are the same. Therefore ΔH can be calculated by adding equations and their enthalpy changes. If an equation is reversed, the sign of ΔH is reversed; if it is multiplied, ΔH is multiplied by the same factor.

Important Conceptual Short Questions — Topic 2

1. Why is the standard enthalpy of formation of oxygen zero?

Oxygen in its standard state, $O_2(g)$, is the reference form of the element. By definition, forming an element from itself involves no enthalpy change.

2. How does Hess's law allow calculation of an enthalpy change that cannot be measured directly?

A target reaction can be expressed as the sum of other reactions with known enthalpy changes. Adding those equations and their ΔH values gives the required ΔH.

3. Why must chemical equations be reversed with a sign change in ΔH?

Reversing a reaction changes the direction of heat flow. If the forward reaction releases heat, the reverse reaction absorbs the same amount.

4. How can combustion data be used to find enthalpy of formation?

Combustion reactions of elements and the compound can be combined using Hess's law so that the overall equation becomes the formation reaction of the compound.

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Topic 3 — Calorimetry, calorie content and bond energy calculations

Q.3 Describe how enthalpy change is measured by calorimetry and calculated from bond energies.

In a glass calorimeter, heat released or absorbed is measured from the temperature change of water: $q = mc\Delta T$. The specific heat capacity of water is 4.18 J g^{-1} K^{-1}, often rounded to 4.2. The enthalpy change per mole is found by dividing q by the number of moles and changing the sign for the reaction.

Bond breaking absorbs energy and bond making releases energy. $\Delta H \approx \Sigma$ bond energies of bonds broken – Σ bond energies of bonds formed. Calorimetry can also estimate the calorie content of foods by measuring the heat released when a food sample is burned.

Textbook data check: Table 6.1 (p. 117) lists the N≡N bond energy as 945 kJ mol^{-1}, while Chapter 12 (p. 266) gives +944 kJ mol^{-1}. Both are rounded textbook values; use the value given in the question.

Important Conceptual Short Questions — Topic 3

1. Why is a calorimeter insulated?

Insulation reduces heat exchange with the surroundings, so the measured temperature change is closer to the heat actually released or absorbed by the reaction.

2. How is q = mcΔT used to find molar enthalpy change?

Calculate the heat absorbed by water, then divide by the moles of reactant that reacted. The reaction's ΔH has the opposite sign to the heat gained by the water.

3. Why does bond breaking require energy?

Energy is needed to overcome the attractive forces holding bonded atoms together. Therefore bond breaking is endothermic.

4. How can bond energy data predict whether a reaction is exothermic?

Add the energies of bonds broken and subtract the energies of bonds formed. If the result is negative, more energy is released than absorbed, so the reaction is exothermic.

Topic 4 — Energetics of solution, lattice energy and Born-Haber cycle

Q.4 Explain hydration energy, lattice energy and the Born-Haber cycle.

When an ionic solid dissolves, the lattice must be broken and ions must be hydrated. Lattice energy is the energy released when one mole of an ionic solid forms from gaseous ions, or the energy required to separate it into gaseous ions. Hydration energy is released when gaseous ions are surrounded by water molecules.

The Born-Haber cycle applies Hess's law to formation of an ionic compound. It combines atomization, ionization, electron affinity and lattice energy. For NaCl, the cycle explains why formation is energetically favourable even though some steps, such as ionization of sodium, require energy. The large lattice energy compensates for these endothermic steps.

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Important Conceptual Short Questions — Topic 4

1. Why does dissolving ammonium nitrate cool the water?

The energy needed to break the ionic lattice is greater than the energy released by hydration. The process absorbs heat from the water, so the temperature falls.

2. How do lattice energy and hydration energy determine whether dissolution is exothermic?

If hydration releases more energy than is needed to break the lattice, dissolution is exothermic. If lattice breaking requires more energy, dissolution is endothermic.

3. Why is the Born-Haber cycle an application of Hess's law?

It calculates lattice energy by adding a series of steps whose total enthalpy change equals the direct formation of the ionic solid.

4. How does ionic charge affect lattice energy?

Higher ionic charges produce stronger electrostatic attraction, so lattice energy becomes more negative. Smaller ionic radii also increase lattice energy.

Topic 5 — Entropy and free energy

Q.5 Explain entropy and Gibbs free energy and their use in predicting spontaneity.

Entropy, S, measures the disorder or number of possible arrangements of particles. Gases have higher entropy than liquids, and liquids higher than solids. Entropy increases when a solid dissolves, when a gas expands or when the number of gas molecules increases.

Gibbs free energy change is $\Delta G = \Delta H - T\Delta S$. A process is spontaneous when ΔG is negative, non-spontaneous when ΔG is positive, and at equilibrium when ΔG is zero. Thus both enthalpy and entropy, together with temperature, determine whether a reaction can occur spontaneously.

Important Conceptual Short Questions — Topic 5

1. Why does melting increase entropy?

Particles in a liquid have more freedom of movement and more possible arrangements than in a solid, so disorder increases.

2. How can an endothermic reaction still be spontaneous?

If the entropy increase is large enough, the term $T\Delta S$ can outweigh the positive ΔH, making ΔG negative.

3. Why does temperature affect the sign of ΔG?

Temperature multiplies the entropy term in $\Delta G = \Delta H - T\Delta S$. At high temperature, the entropy contribution becomes more important.

4. How does ΔG = 0 relate to equilibrium?

When $\Delta G = 0$, there is no net driving force for the forward or reverse reaction. The system is at equilibrium.

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